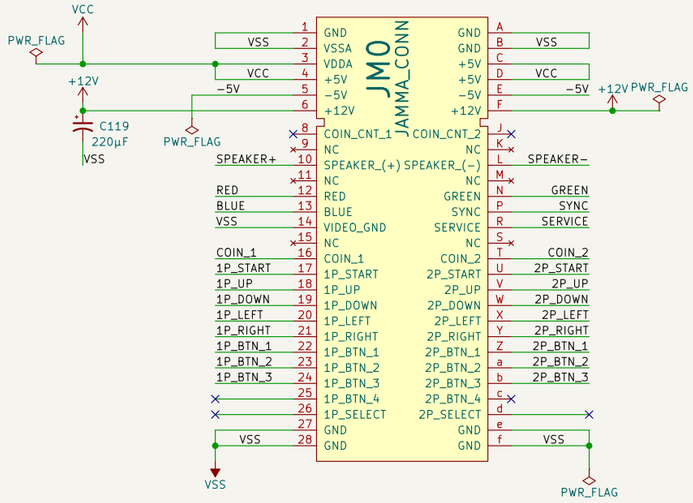
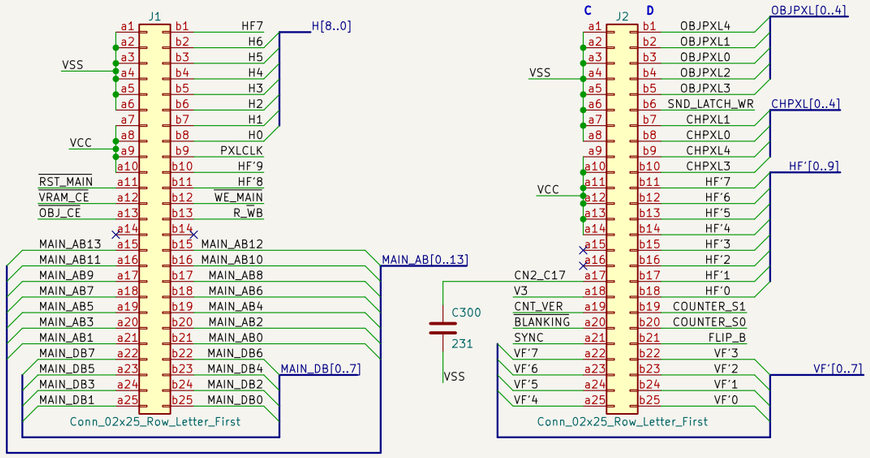
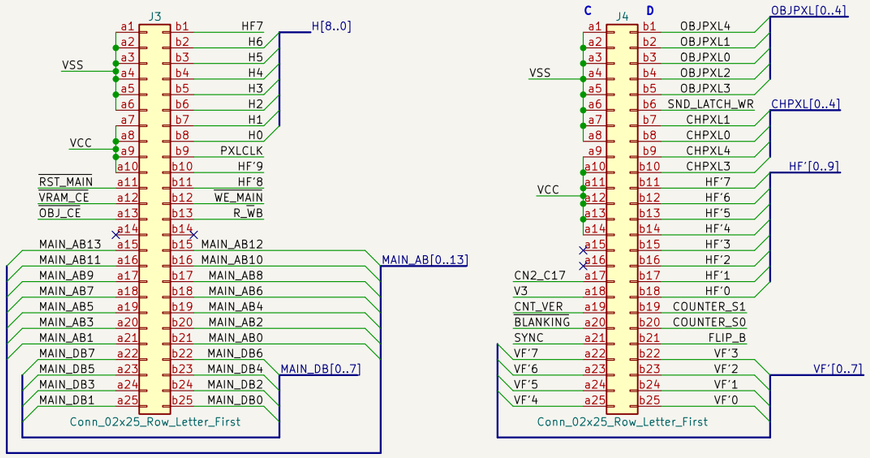
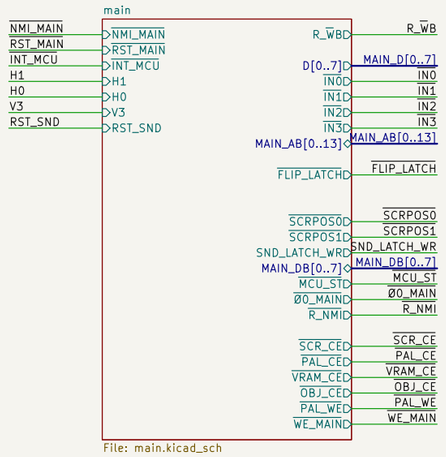


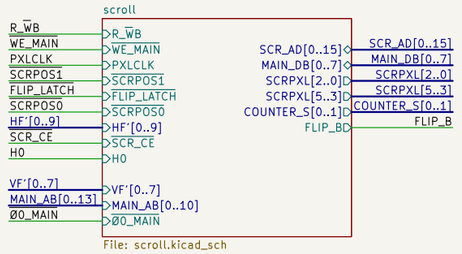
Top board connectors



Bottom board connectors



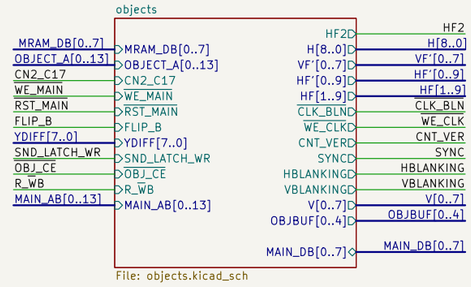


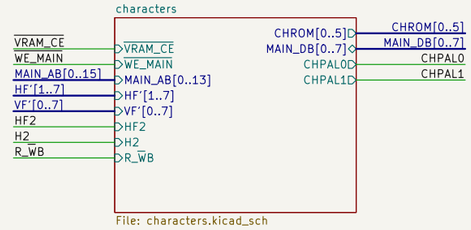


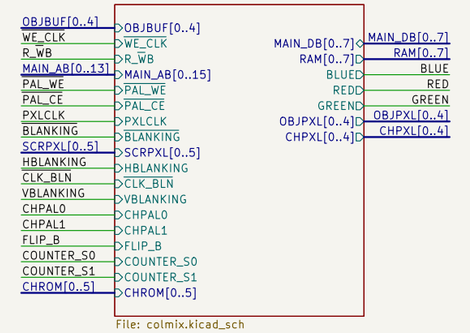


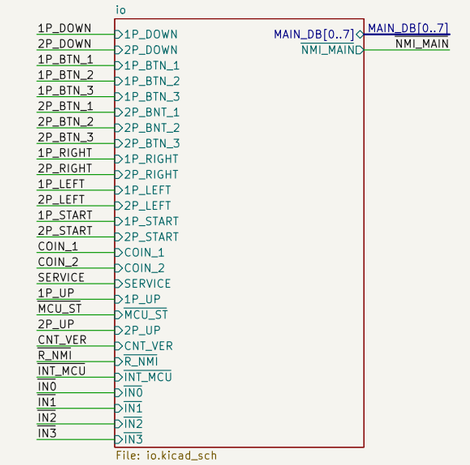












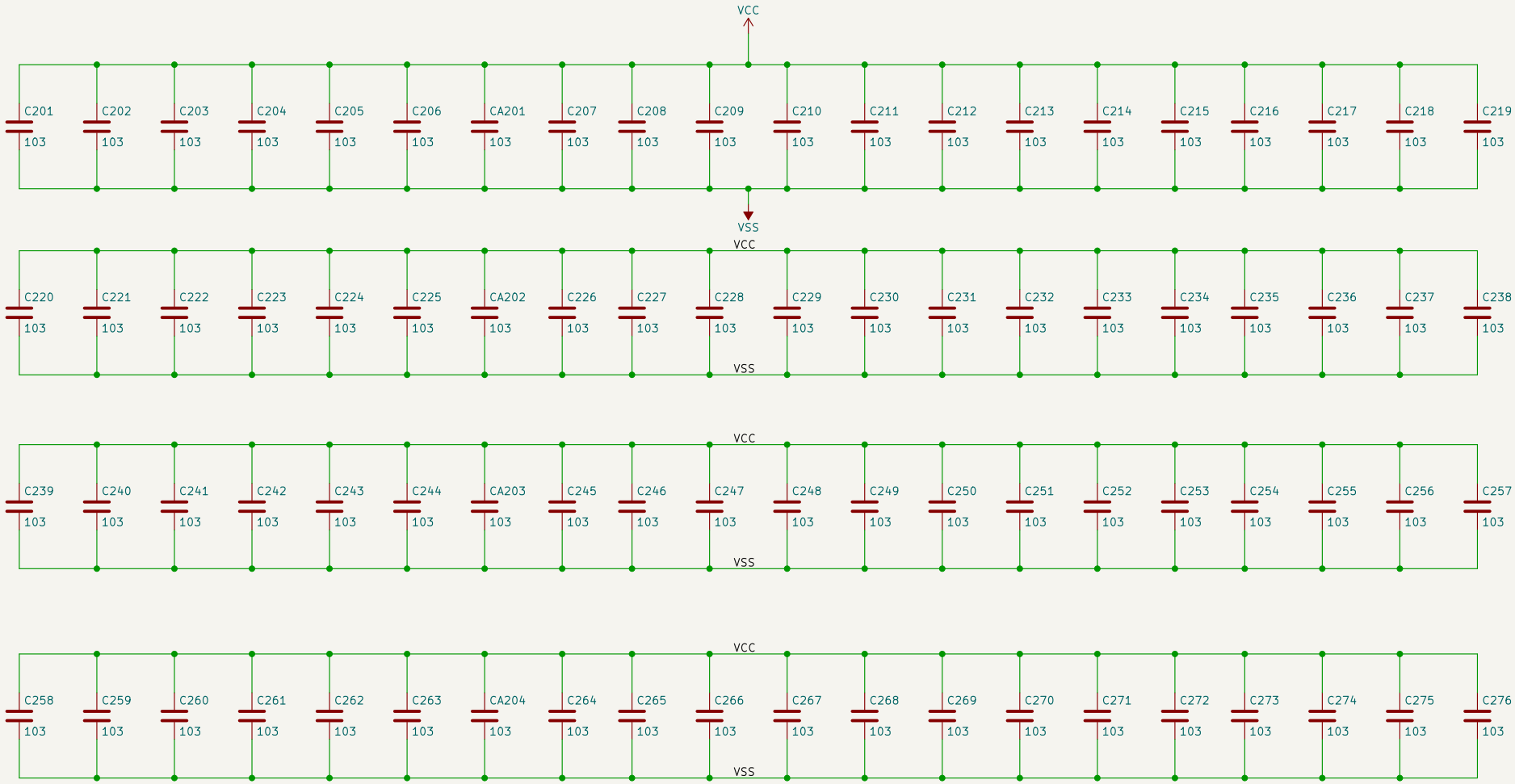
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Sheet: /
File: kunio.kicad_sch

Title: RENEGADE

Size: A3	Date: 2022-07-08	**Rev: José Tejada**
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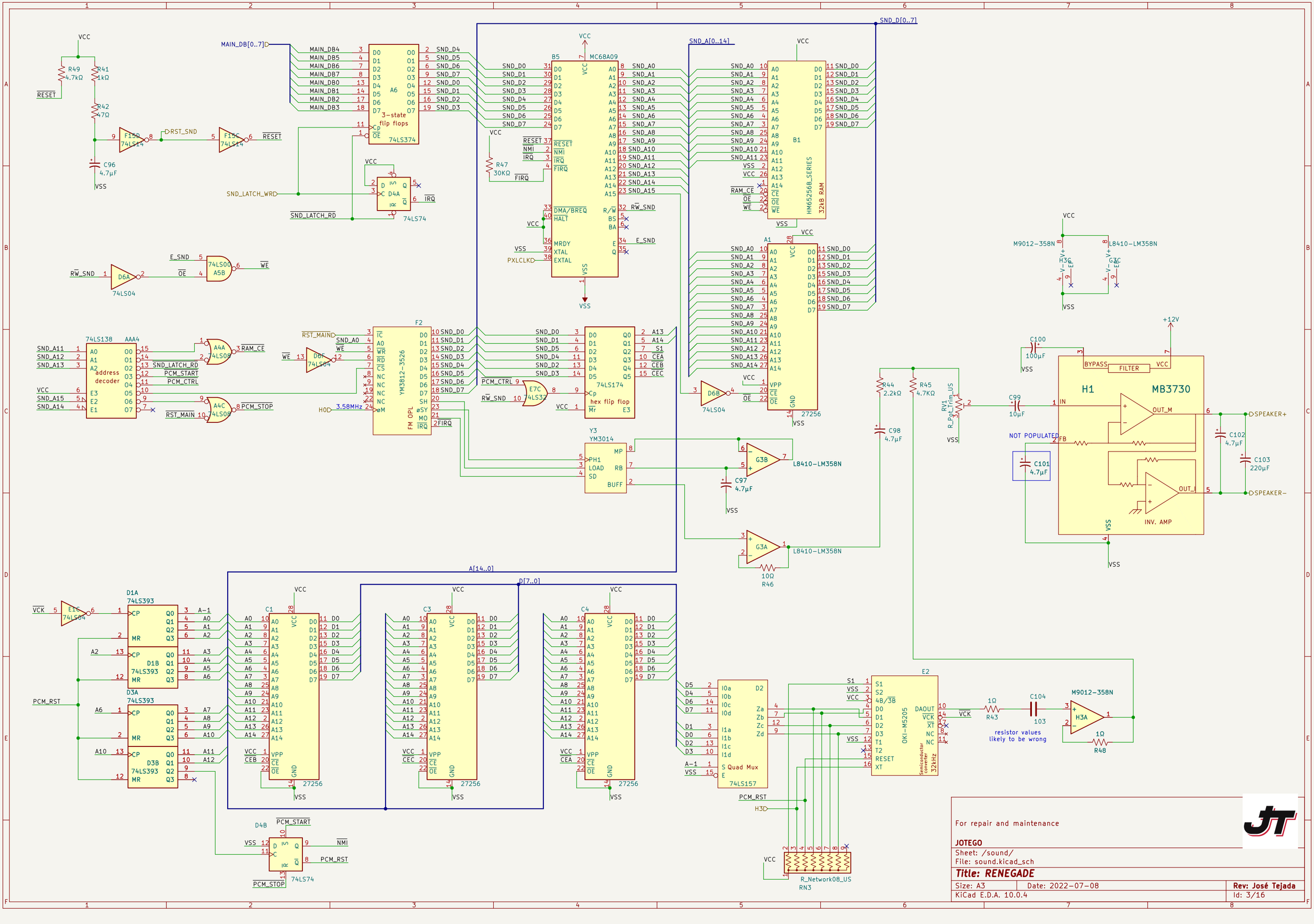
For repair and maintenance

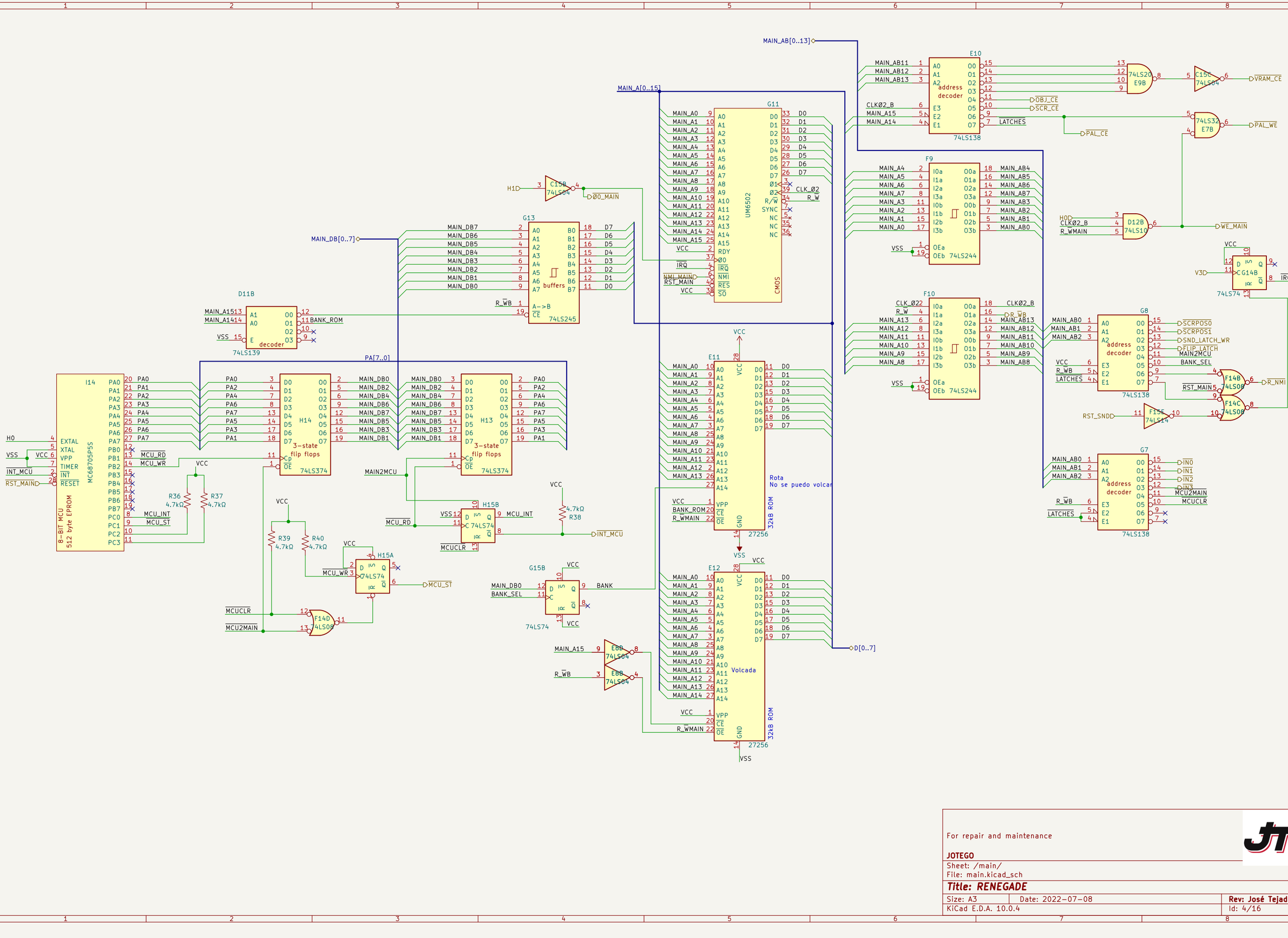
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Title: RENEGADE

Size: A4	Date: 2022-07-08	**Rev: José Tejada**
KiCad E.D.A. 10.0.4		Id: 2/16


SND_D[0..7]
SND_A[0..14]
MAIN_DB[0..7]
A[14..0]
D[7..0]
VCC
VSS
RESET
R49 4.7kΩ
R41 1kΩ
R42 47Ω
C96 4.7µF
F15D 74LS14
F15C 74LS14
RST_SND
A6 3-state flip flops 74LS374
MAIN_DB4
MAIN_DB5
MAIN_DB6
MAIN_DB7
MAIN_DB0
MAIN_DB1
MAIN_DB2
MAIN_DB3
SND_D4
SND_D5
SND_D6
SND_D7
SND_D0
SND_D1
SND_D2
SND_D3
SND_LATCH_WR
SND_LATCH_RD
B4 D FF 74LS74
IRQ
B5 MC68A09
SND_A0
SND_A1
SND_A2
SND_A3
SND_A4
SND_A5
SND_A6
SND_A7
SND_A8
SND_A9
SND_A10
SND_A11
SND_A12
SND_A13
SND_A14
SND_A15
RESET
NMI
IRQ
FIRQ
R47 30KΩ
DMA/BREQ
HALT
MRDY
XTAL
EXTAL
PXLCLK
R/W
RW_SND
BS
BA
E_SND
Q
B1 HM65256B-SERIES 32kB RAM
RAM_CE
OE
WE
A1 27256
D6B 74LS04
M9012-358N
L8410-LM358N
R36
R37
C100 100µF
+12V
H1 MB3730
BYPASS
FILTER
IN
OUT_M
OUT_I
INV. AMP
SPEAKER+
SPEAKER-
NOT POPULATED
C101 4.7µF
C102 4.7µF
C103 220µF
R44 2.2kΩ
R45 4.7kΩ
RV1 R_Pot_Trim_US
C99 10µF
C98 4.7µF
74LS138 AAA4
address decoder
A4A 74LS08
A4C 74LS08
RAM_CE
SND_LATCH_RD
PCM_START
PCM_CTRL
PCM_STOP
RST_MAIN
PCM_RST
F2 YM3812-3526 FM OPL
H0
3.58MHz
D6F 74LS04
E7C 74LS32
C97 4.7µF
Y3 YM3014
MP
PH1
LOAD
RB
SD
BUFF
G3B
G3A
10Ω R46
C104
103
1Ω R43
1Ω R48
resistor values likely to be wrong
E2 OKI-M5205 Semiconductor converter 32kHz
S1
S2
4B/3B
DAOUT
VCK
XT
RESET
D1A D1B D3A D3B 74LS393
E1C 74LS04
C1
C3
C4
D4B 74LS74
E_SND
RW_SND
D6A 74LS04
74LS00 A5B
WE
A13
A14
CEA
CEB
CEC
74LS174 hex flip flop
74LS157 S Quad Mux
R_Network08_US RN3
For repair and maintenance
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Title: RENEGADE
Size: A3
Date: 2022-07-08
Rev: José Tejada
KiCad E.D.A. 10.0.4
Id: 3/16



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JOTEGO

Sheet: /main/
File: main.kicad_sch

Title: RENEGADE

Size: A3	Date: 2022-07-08	**Rev: José Tejada**
KiCad E.D.A. 10.0.4		Id: 4/16

MAIN_AB[0..10]
MAIN_AB0
MAIN_AB1
MAIN_AB3
MAIN_AB2
SCR_CA0
SCR_CA3
SCR_CA2
SCR_CA1
SCR_CE
VSS
74LS157
SCR_A0
SCR_A1
SCR_A3
SCR_A2
SCR_A[9..0]
RAM_2016 C6
SCR_AD0
SCR_AD1
SCR_AD2
SCR_AD3
SCR_AD4
SCR_AD5
SCR_AD6
SCR_AD7
lower half
CS_SCR
WE_SCR
SCR_AD[0..15]
B7
buffers
74LS245
R_WB
RD_SND
74LS04
MAIN_DB[0..7]
MAIN_DB0
MAIN_DB1
MAIN_DB2
MAIN_DB3
MAIN_DB4
MAIN_DB5
MAIN_DB6
MAIN_DB7
74LS174
hex flip flop
SCRPOS1
SCRPOS9
SCRPOS8
74LS273
flip flops
SCRPOS0
SCRPOS4
SCRPOS5
SCRPOS6
SCRPOS7
SCRPOS0
SCRPOS1
SCRPOS2
SCRPOS3
Horizontal
Counnter
HF'[0..9]
74LS283
SUM
C93
221
PXLCLK
REPAIR
CARRY0
NOT ORIGINALLY CONNECTED
CONNECTION REPAIRED
SD0
SD1
SD2
SD3
SD4
SD5
SD6
SD7
SD[7..0]
SCR_CA0
SCR_CA1
SCR_CA2
SCR_CA3
SCROM_A4
SCR_CA5
SCR_CA6
SCR_CA7
REPAIR
CARRY0
VF'5
VF'4
MAIN_AB4
MAIN_AB5
MAIN_AB7
MAIN_AB6
SCR_A4
SCR_A5
SCR_A7
SCR_A6
upper half
WE_MAIN
MAIN_AB10
WE_SCR
CE_SCR
RD_SND
VF'6
VF'7
SCR_A8
SCR_A9
B8
B10
B11
F7
C15E
CLK_SCOLM
SCR0
SCR1
SCR2
SCR3
SCR4
SCR5
SCR6
SCR7
SCR8
SCR9
SCR10
SCR13
SCR14
SCR15
SCR[15..0]
SCROM_A[15..0]
SCROM_A13
SCROM_A12
SCROM_A11
SCROM_A10
SCROM_A9
SCROM_A8
SCROM_A7
SCROM_A6
SCROM_A14
SCROM_A15
CE_SCROM
SCOL0
SCOL1
SCOL2
SCRPXL3
SCRPXL4
SCRPXL5
SCRPXL[5..3]
Ø0_MAIN
D8
D9
F8
VF'[0..7]
scrom_memories
SCROM_A[15..0]
PXLCLK
MAIN_DB0
FLIP_LATCH
H0
VF'[0..3]
CE_SCROM
SCR_CA5
SCR_CA6
SCR_CA7
SCRPXL[2..0]
COUNTER_S1D
COUNTER_S0D
CLK_SCOLMD
FLIP_BD
COUNTER_S1
COUNTER_S0
CLK_SCOLM
FLIP_B
COUNTER_S[0..1]
File: scrom_memories.kicad_sch

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Sheet: /scroll/
File: scroll.kicad_sch

Title: RENEGADE

Size: A3 | Date: 2022-07-08 | Rev: José Tejada
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SCROM_A[15..0]

SCR_CA5 — C15F 74LS04 — SCROM_A5

CE_SCROM — C15A 74LS04 — CE_SCRB

D11A 74LS139 decoder
CE_SCROM, SCROM_A14 → A1, A0
VSS → E
O0 CE_SCRF
O1 CE_SCRD
O2 CE_SCRE
O3 CE_SCRC

VF'[0..3]

A15 27256
VF'0–VF'3 → A0–A3
SCROM_A4–SCROM_A14 → A4–A14
VCC → VCC, CE_SCROM → CE, VSS → GND
D0 SCROM_D0
D1 SCROM_D1
D2 SCROM_D2
D3 SCROM_D3
D4 SCROM_D4
D5 SCROM_D5
D6 SCROM_D6
D7 SCROM_D7

B15 27256
VF'0–VF'3 → A0–A3
SCROM_A4–SCROM_A14 → A4–A14
VCC → VCC, CE_SCRB → CE, VSS → GND
D0 SCROM_D0
D1 SCROM_D1
D2 SCROM_D2
D3 SCROM_D3
D4 SCROM_D4
D5 SCROM_D5
D6 SCROM_D6
D7 SCROM_D7

SCROM_D[0..15]

SCROM_D0, SCROM_D1, SCROM_D3, SCROM_D2 → I0a, I0b, I0c, I0d
SCROM_D4, SCROM_D5, SCROM_D7, SCROM_D6 → I1a, I1b, I1c, I1d
SCROM_A15 → S
VSS → E
Za 4
Zb 7
Zc 12
Zd 9
Quad Mux 74LS157

C14 27256
VF'0–VF'3 → A0–A3
SCROM_A4–SCROM_A15 → A4–A15
VCC → VCC, CE_SCRC → CE, VSS → GND
D0 SCROM_D8
D1 SCROM_D9
D2 SCROM_D10
D3 SCROM_D11
D4 SCROM_D12
D5 SCROM_D13
D6 SCROM_D14
D7 SCROM_D15

C11 27256
VF'0–VF'3 → A0–A3
SCROM_A4–SCROM_A15 → A4–A15
VCC → VCC, CE_SCRD → CE, VSS → GND
D0 SCROM_D8
D1 SCROM_D9
D2 SCROM_D10
D3 SCROM_D11
D4 SCROM_D12
D5 SCROM_D13
D6 SCROM_D14
D7 SCROM_D15

C12 27256
VF'0–VF'3 → A0–A3
SCROM_A4–SCROM_A15 → A4–A15
VCC → VCC, CE_SCRE → CE, VSS → GND
D0 SCROM_D8
D1 SCROM_D9
D2 SCROM_D10
D3 SCROM_D11
D4 SCROM_D12
D5 SCROM_D13
D6 SCROM_D14
D7 SCROM_D15

C9 27256
VF'0–VF'3 → A0–A3
SCROM_A4–SCROM_A15 → A4–A15
VCC → VCC, CE_SCRF → CE, VSS → GND
D0 SCROM_D8
D1 SCROM_D9
D2 SCROM_D10
D3 SCROM_D11
D4 SCROM_D12
D5 SCROM_D13
D6 SCROM_D14
D7 SCROM_D15

SR0–1, SR0–0 → S0, S1
PXLCLKD → Cp
VCC → CL
Dsl VSS
P3 SCROM_D15, P2 SCROM_D14, P1 SCROM_D13, P0 E14 SCROM_D12
Q0 15, Q1 14, Q2 13, Q3 12, Dsr 2 VSS
shift register 74LS194

SR0–1, SR0–0 → S0, S1
PXLCLK → Cp
VCC → CL
Dsl VSS
P3 SCROM_D11, P2 SCROM_D10, P1 SCROM_D9, P0 D14 SCROM_D8
Q0 15, Q1 14, Q2 13, Q3 12, Dsr 2 VSS
shift register 74LS194

I0a E15, I0b, I0c, I0d 14 11
I1a 3, I1b 6, I1c 13, I1d 10
FLIP → S, VSS → E
Za 4 SCRPXL1
Zb 7 SCRPXL0
Zc 12 SCRPXL2
Zd 9
Quad Mux 74LS157
SCRPXL[2..0]

SCR_CA6 → D13B 74LS86 (4, 5, 6)
SCR_CA7 → D13A 74LS86 (1, 2, 3)
D13B → F14A 74LS08 (1, 2, 3)
H0 → F14A
VCC → G15A
MAIN_DB0 → D
FLIP_LATCHD → C
G15A 74LS74 (D, C, Q, ~Q)
FLIP → D13D 74LS86 (12, 13, 11)
SCROM_A4 → D13D
D13D → CLK_SCOLM

I0a D15, I0b, I0c, I0d 14 11
I1a 3, I1b 6, I1c 13, I1d 10
FLIP → S, VSS → E
Za 4 COUNTER_S0
Zb 7 SR0–0
Zc 12 COUNTER_S1
Zd 9 SR0–1
Quad Mux 74LS157

G15A Q → D6E 74LS04 (11, 10) → FLIP_B

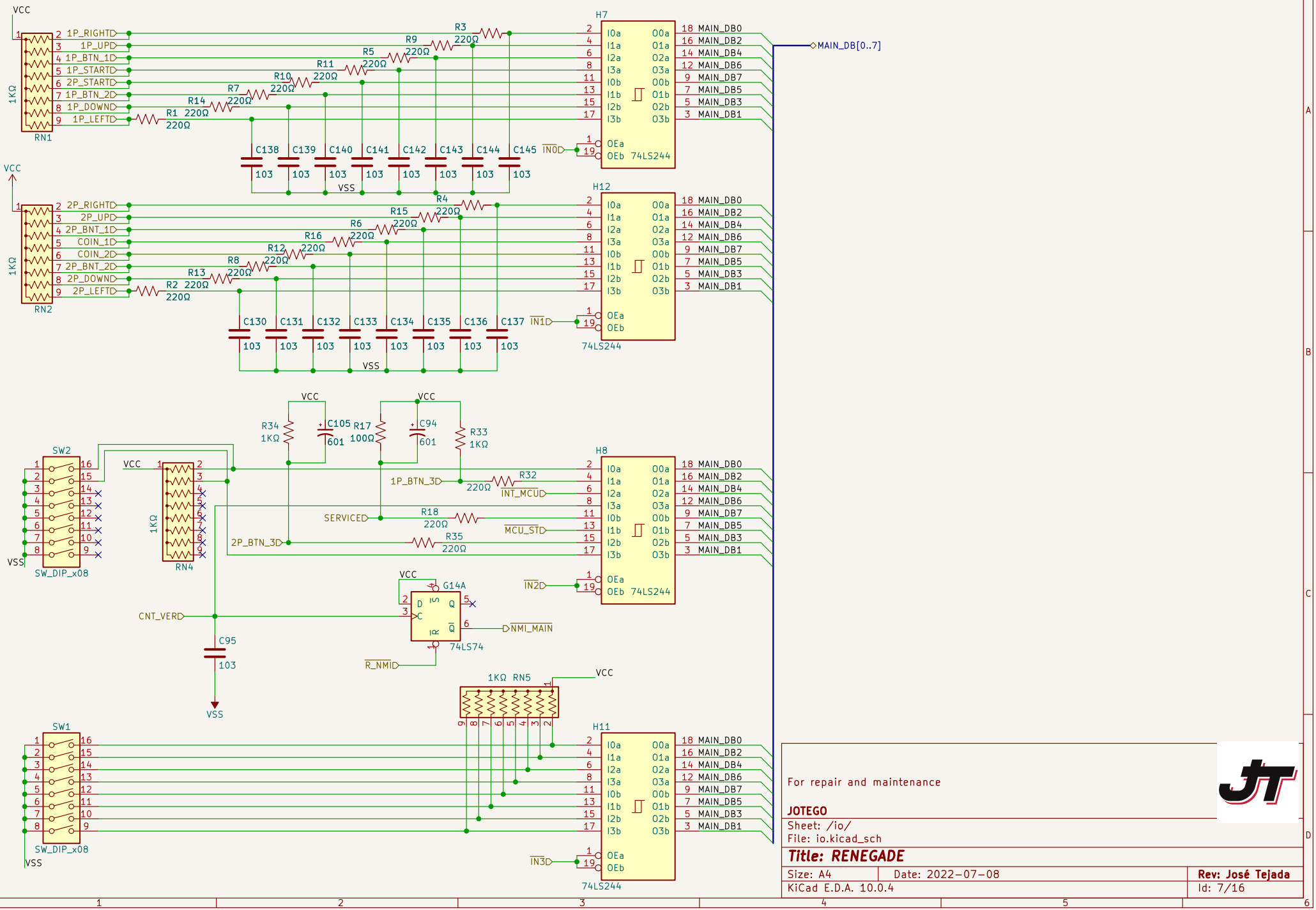
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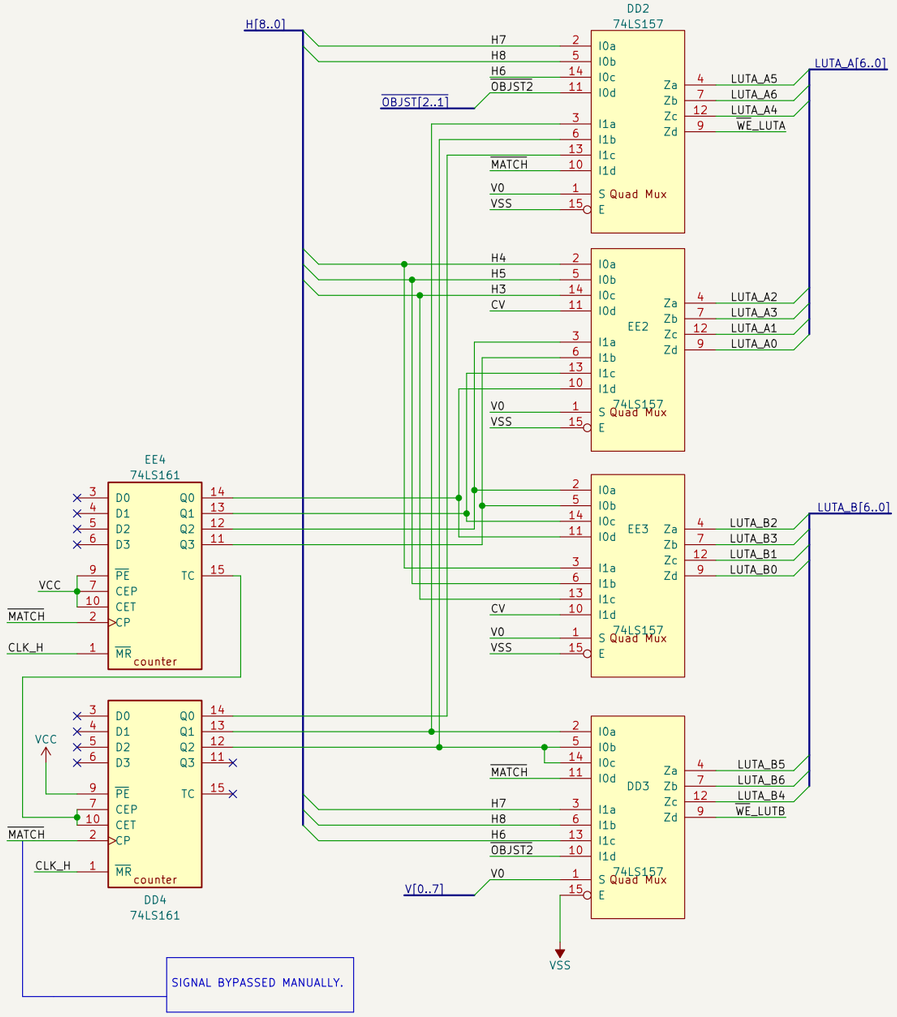
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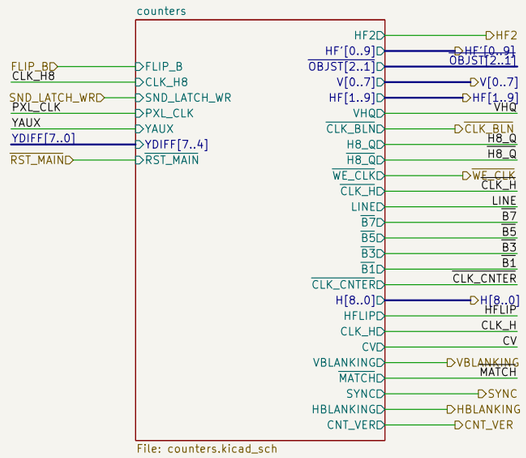
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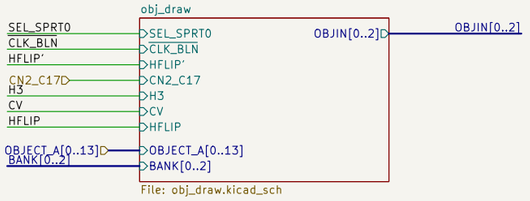
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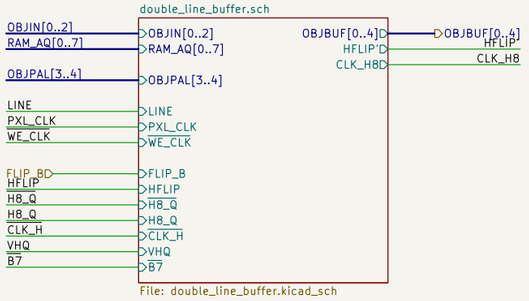
Size: A3	Date: 2022-07-08	**Rev: José Tejada**
KiCad E.D.A. 10.0.4		Id: 6/16

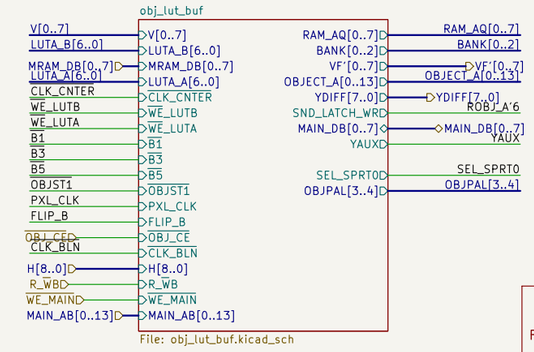

VCC
1P_RIGHT
1P_UP
1P_BTN_1D
1P_START
2P_START
1P_BTN_2D
1P_DOWN
1P_LEFT
RN1
1KΩ
R1 220Ω
R2 220Ω
R3 220Ω
R4 220Ω
R5 220Ω
R6 220Ω
R7 220Ω
R8 220Ω
R9 220Ω
R10 220Ω
R11 220Ω
R12 220Ω
R13 220Ω
R14 220Ω
R15 220Ω
R16 220Ω
C138 C139 C140 C141 C142 C143 C144 C145
C130 C131 C132 C133 C134 C135 C136 C137
103
VSS
IN0D
IN1D
H7
H12
74LS244
I0a I1a I2a I3a I0b I1b I2b I3b
O0a O1a O2a O3a O0b O1b O2b O3b
OEa OEb
MAIN_DB0
MAIN_DB2
MAIN_DB4
MAIN_DB6
MAIN_DB7
MAIN_DB5
MAIN_DB3
MAIN_DB1
MAIN_DB[0..7]
2P_RIGHT
2P_UP
2P_BNT_1D
COIN_1D
COIN_2D
2P_BNT_2D
2P_DOWN
2P_LEFT
RN2
SW2
SW_DIP_x08
RN4
R34 1KΩ
R17 100Ω
R33 1KΩ
C105 601
C94 601
R32 220Ω
R18 220Ω
R35 220Ω
1P_BTN_3D
2P_BTN_3D
SERVICED
INT_MCU
MCU_ST
IN2D
H8
G14A
74LS74
CNT_VER
C95
NMI_MAIN
R_NMI
1KΩ RN5
SW1
IN3D
H11
For repair and maintenance
JOTEGO
Sheet: /io/
File: io.kicad_sch
Title: RENEGADE
Size: A4
Date: 2022-07-08
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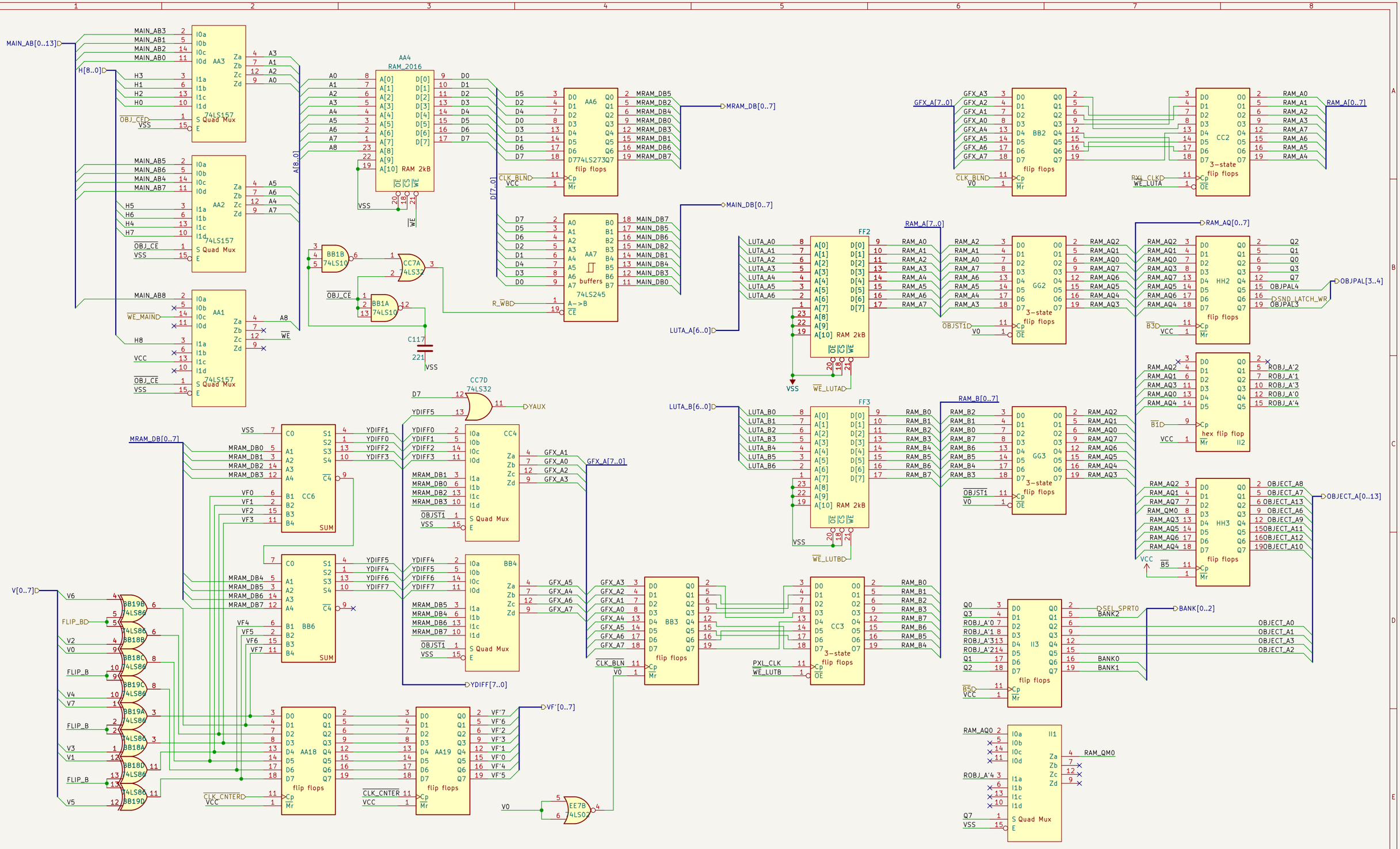
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JOTEGO

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Title: RENEGADE

Size: A3	Date: 2022-07-08	**Rev: José Tejada**
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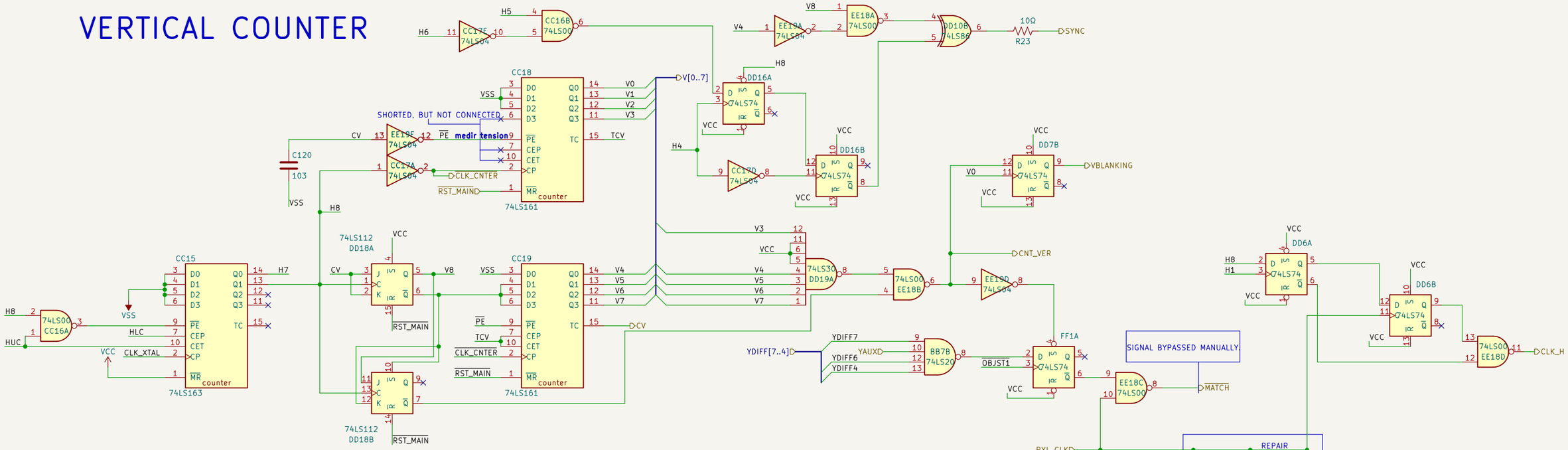
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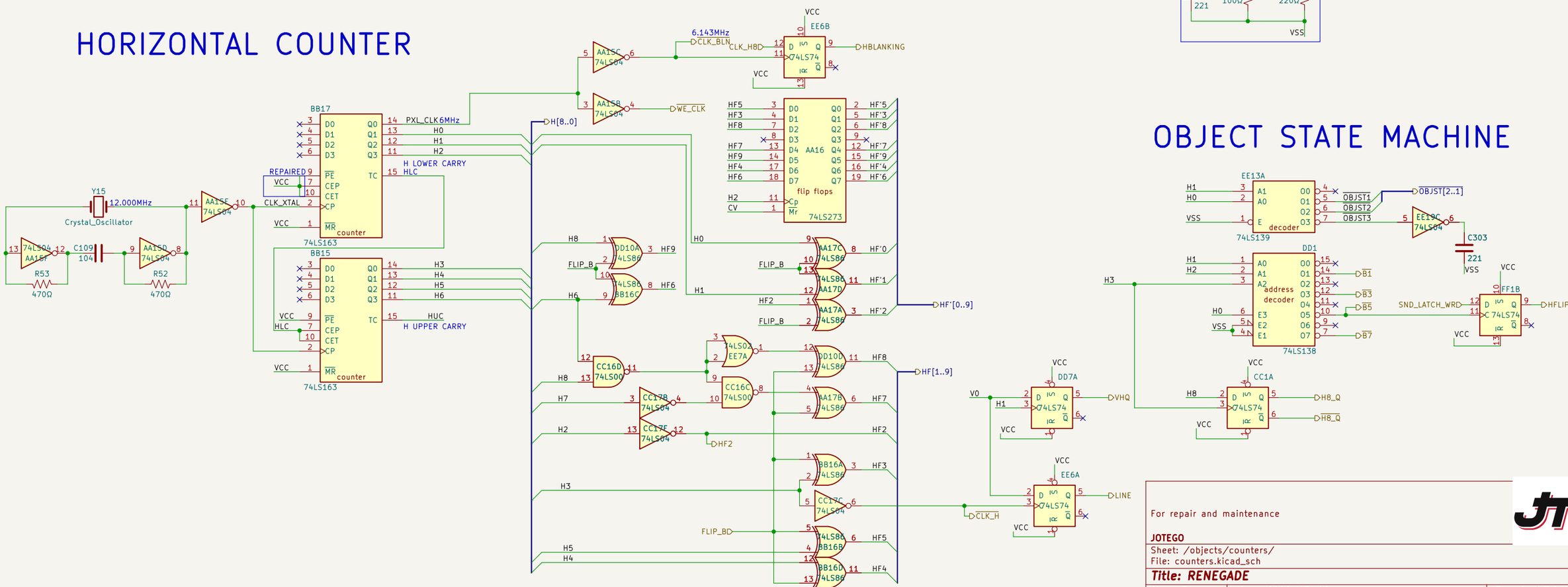
Title: RENEGADE

Size: A3	Date: 2022-07-08	**Rev: José Tejada**
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VERTICAL COUNTER



HORIZONTAL COUNTER



OBJECT STATE MACHINE

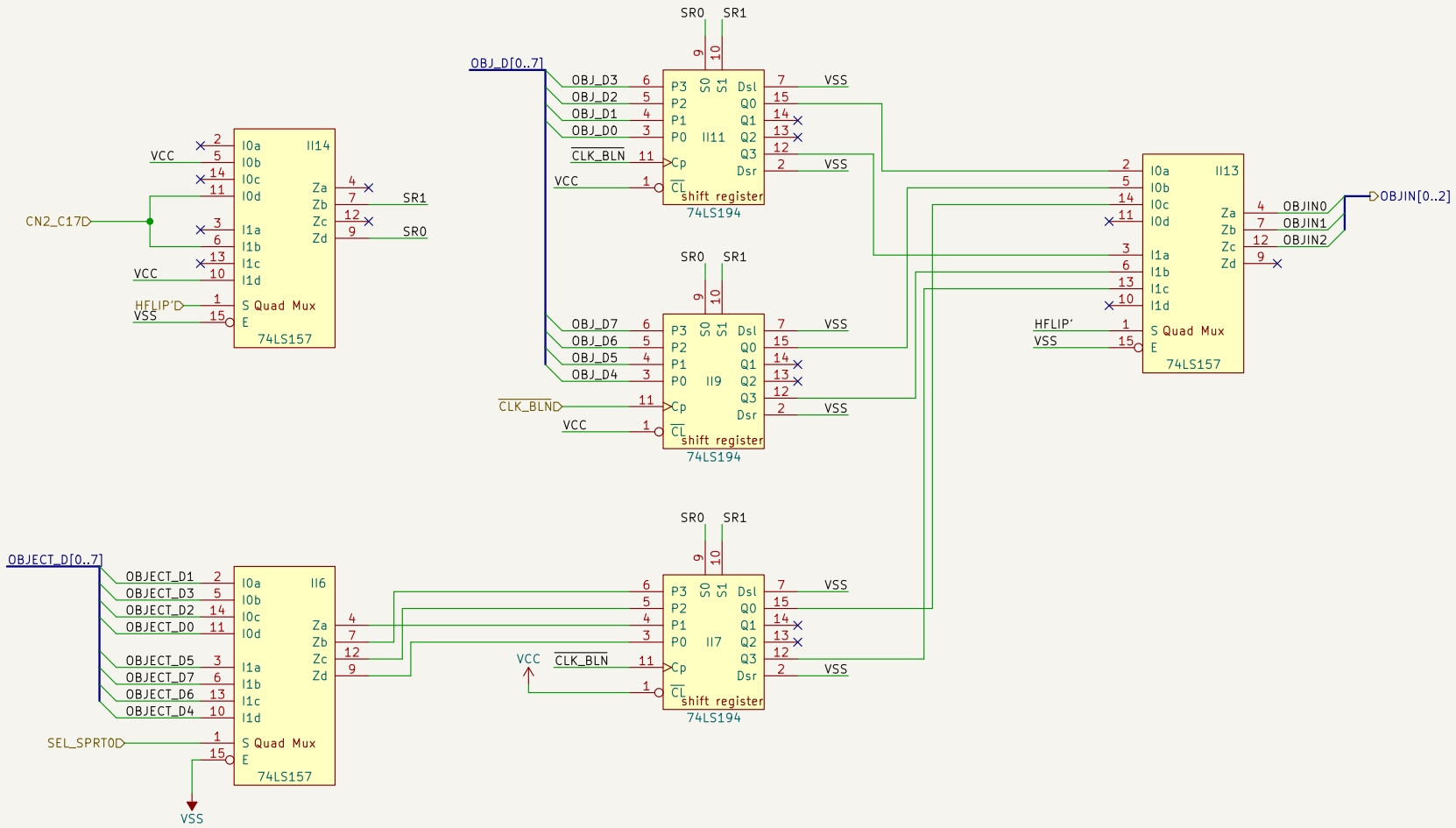
For repair and maintenance

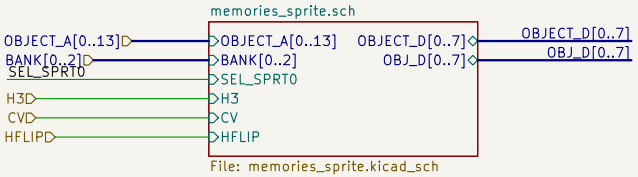
JOTEGO

Sheet: /objects/counters/
File: counters.kicad_sch

Title: RENEGADE

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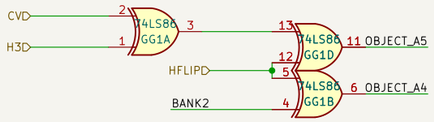
For repair and maintenance

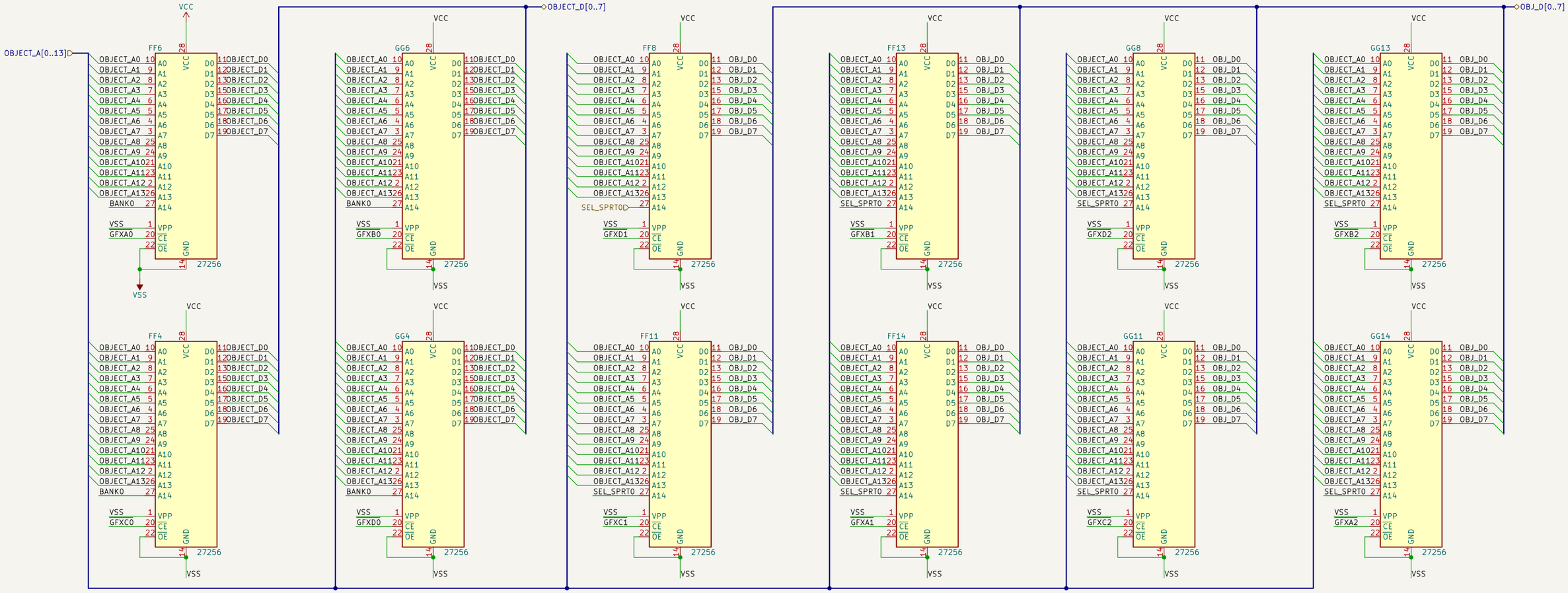
JOTEGO

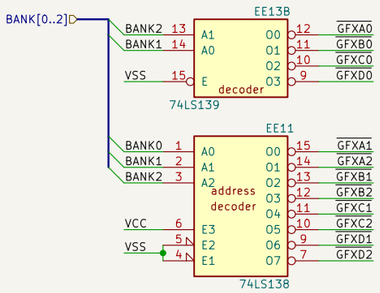
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File: obj_draw.kicad_sch

Title: RENEGADE

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Memorias volcadas

For repair and maintenance

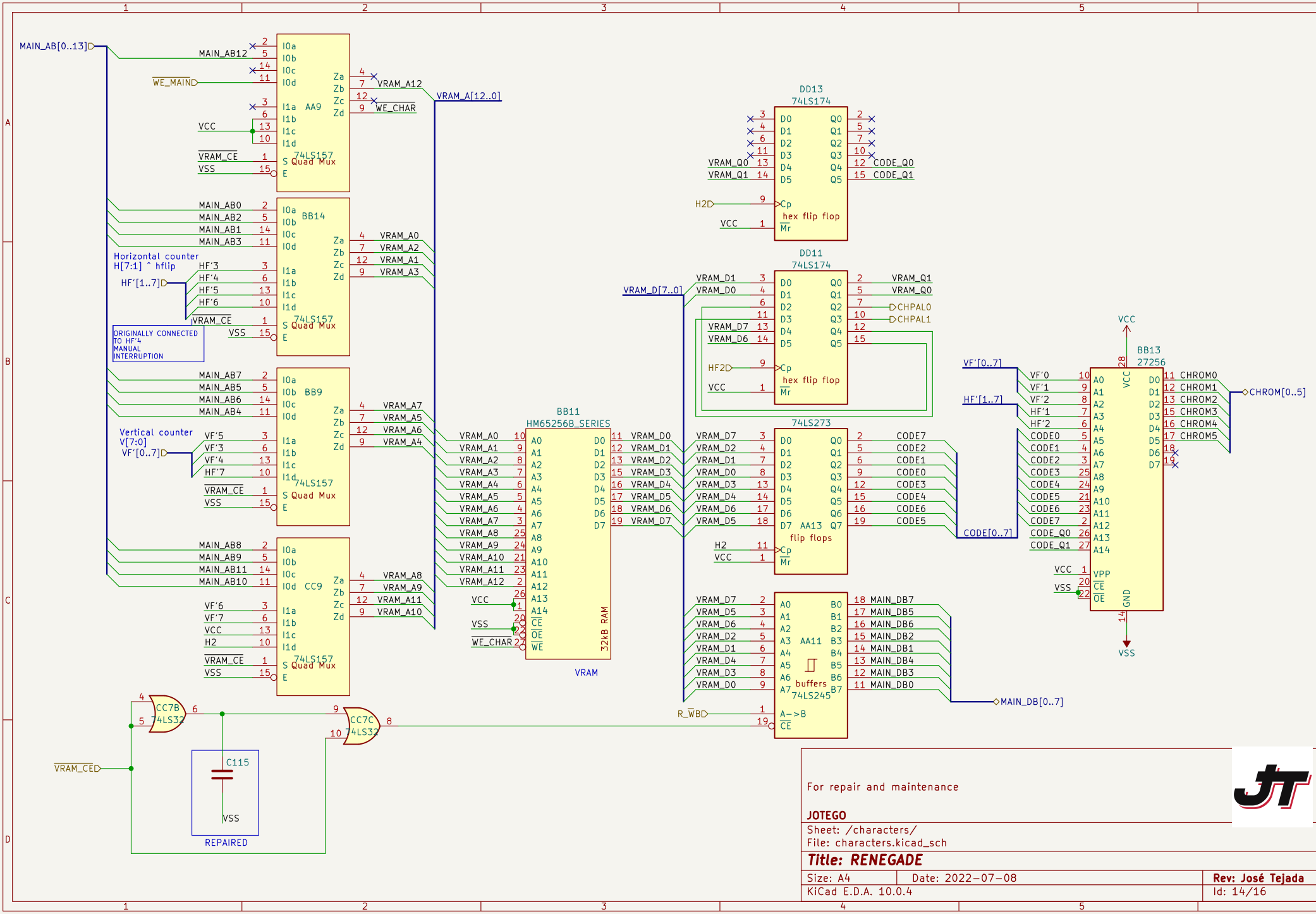
JOTEGO

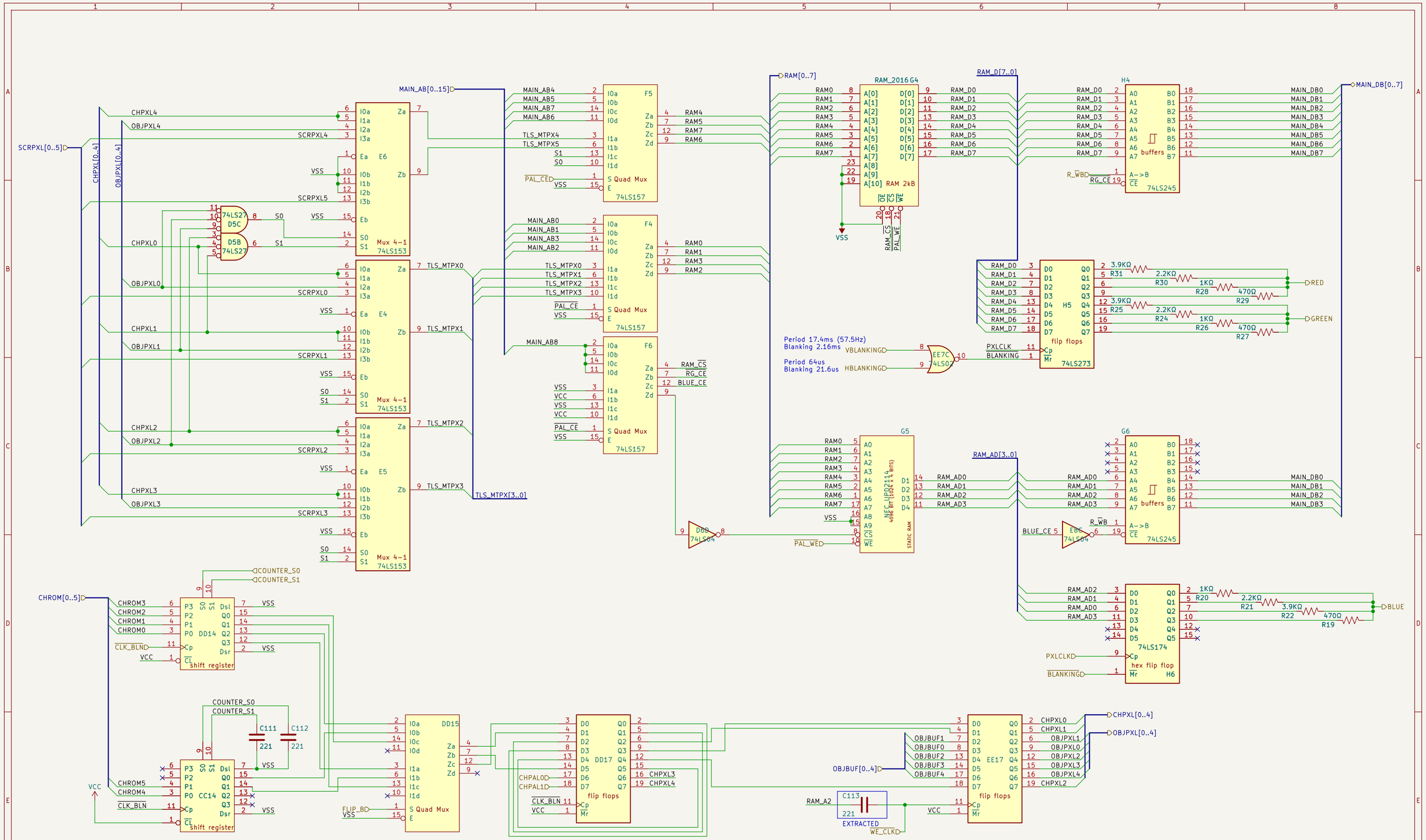
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File: memories_sprite.kicad_sch

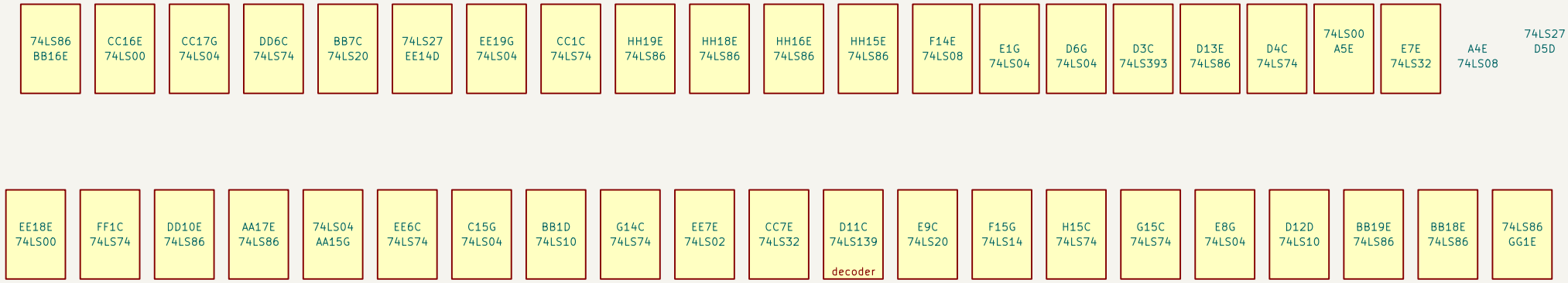
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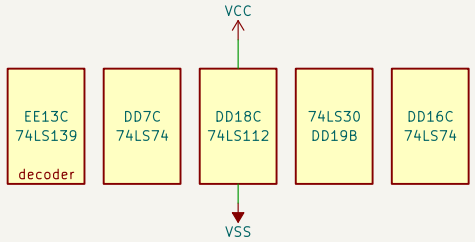
Size: A3	Date: 2022-07-08	**Rev: José Tejada**
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RAM_AQ[0..7]
II18
RAM_AQ4
RAM_AQ5
RAM_AQ6
RAM_AQ7
PE_LINEA
WE_CLK
MR_A
D0
D1
D2
D3
PE
CEP
CET
CP
MR
Q0
Q1
Q2
Q3
TC
counter
74LS163
II19
RAM_AQ0
RAM_AQ1
RAM_AQ2
RAM_AQ3
C118
100
VCC
II15
RAM_AQ4
RAM_AQ5
RAM_AQ6
RAM_AQ7
PE_LINEB
MR_B
II16
RAM_AQ0
RAM_AQ1
RAM_AQ2
RAM_AQ3
PE_LINEB
VCC
WE_CLK
MR_B
VSS
FLIP_BD
LINE
S Quad Mux
E
74LS157
HH17
I0a
I0b
I0c
I0d
I1a
I1b
I1c
I1d
Za
Zb
Zc
Zd
HH19C
HH19D
HH19A
HH19B
HH18C
HH18D
HH18A
HH18B
HH16C
HH16D
HH16A
HH16B
HH15C
HH15D
HH15A
HH15B
74LS86
RAM_2016 GG18
RAM_2016 GG16
A[0]
A[1]
A[2]
A[3]
A[4]
A[5]
A[6]
A[7]
A[8]
A[9]
A[10]
RAM 2kB
D[0]
D[1]
D[2]
D[3]
D[4]
D[5]
D[6]
D[7]
OE
CS
WE
WE_CLK
LINEA0
LINEA1
LINEA2
LINEA3
LINEA4
LINEB0
LINEB1
LINEB2
LINEB3
LINEB4
LINEA[4..0]
LINEB[4..0]
OBJIN[0..2]
OBJIN0
OBJIN1
OBJIN2
BLANK
BLANK'
BUFIN0
BUFIN1
BUFIN2
BUFIN3
BUFIN4
HFLIP
HFLIP'
LINED
EE15
EE16
EE14B
74LS27
74LS157
OBJPAL[3..4]
OBJPAL3
OBJPAL4
PXL_CLK
VCC
hex flip flop
Mr
FF18
FF16
FF15
FF17
HH1
74LS174
OBJBUF[0..4]
OBJBUF0
OBJBUF1
OBJBUF2
OBJBUF3
OBJBUF4
GG19
GG17
GG15
74LS257
R54
330Ω
REPAIRED
CC1B
H8_Q
CLK_H
CLK_H8
74LS74
EE13B
74LS04
BB1C
74LS10
II17
VHQ
C301
101
B7
For repair and maintenance
JOTEGO
Sheet: /objects/double_line_buffer.sch/
File: double_line_buffer.kicad_sch
Title: RENEGADE
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MAIN_AB[0..13]
AA9
BB14
BB9
CC9
74LS157
Quad Mux
VRAM_A[12..0]
Horizontal counter
H[7:1] ^ hflip
HF'[1..7]
ORIGINALLY CONNECTED TO HF'4 MANUAL INTERRUPTION
Vertical counter
V[7:0]
VF'[0..7]
BB11
HM65256B_SERIES
32kB RAM
VRAM
DD13
74LS174
hex flip flop
DD11
CHPAL0
CHPAL1
74LS273
flip flops
AA13
AA11
buffers
74LS245
A->B
MAIN_DB[0..7]
BB13
27256
CHROM[0..5]
CC7B
CC7C
74LS32
C115
REPAIRED
For repair and maintenance
JOTEGO
Sheet: /characters/
File: characters.kicad_sch
Title: RENEGADE
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SCRPXL[0..5]
CHPXL[0..4]
OBJPXL[0..4]
MAIN_AB[0..15]
RAM[0..7]
RAM_D[7..0]
MAIN_DB[0..7]
RAM_AD[3..0]
TLS_MTPX[3..0]
CHROM[0..5]
COUNTER_S0
COUNTER_S1
Mux 4-1 74LS153
74LS157 Quad Mux
RAM_2016 G4
RAM 2kB
buffers 74LS245
flip flops 74LS273
hex flip flop 74LS174
shift register
Period 17.4ms (57.5Hz)
Blanking 2.16ms
Period 64us
Blanking 21.6us
VBLANKING
HBLANKING
RED
GREEN
BLUE
For repair and maintenance
JOTEGO
Sheet: /colmix/
File: colmix.kicad_sch
Title: RENEGADE
Size: A3
Date: 2022-07-08
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JOTEGO

Sheet: /last unit/
File: last unit.kicad_sch

Title: RENEGADE

Size: A4	Date: 2022-07-08	**Rev: José Tejada**
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